CHA-2.D.1 The derivative at a point can be estimated from information given in tables or graphs.

CHA-2.D.2 Technology can be used to calculate or estimate the value of a derivative of a function at a point.

2-3 Estimating Derivatives of a Function at a Point

Recall…

The Derivative of a Function tells us …	• The (instantaneous) rate of change at a point • The slope of the tangent line to the curve at a point

Example 1: Explain the meaning of $f'(3) = -7$

The function $f(x)$ is changing at a rate of -7 at $x = 3$.

Approximating/Estimating Derivative Values at a point	
Given	**Method**
Table of Values	Use two points close to the point and a difference quotient (AROC)
Graph of $f(x)$	Calculate slope of a tangent line to the graph at the point
Equation of $f(x)$ and a calculator	Use the derivative function of calculator

Approximating a Derivative with TABLES

x	1	3	7	9
$f(x)$	-2	1	5	3

Example 2: The function $f(x)$ is continuous and differentiable for all x with selected values given above.

a. Approximate $f'(2)$

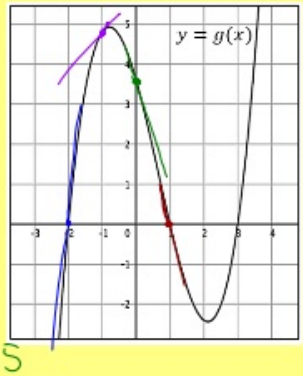
$$f'(2) \approx \frac{f(3)-f(1)}{3-1} = \frac{1--2}{3-1} = \frac{3}{2}$$

b. Estimate $f'(5)$

$$f'(5) \approx \frac{f(7)-f(3)}{7-3} = \frac{5-1}{4} = 1$$

Approximating a Derivative with GRAPHS

Example 3: Use the graph of $g(x)$ at the right to approximate the following values.



a. $g'(-2) \approx \frac{2}{.2} = 10$

b. $g'(-1) \approx \frac{5-4}{-0.8-1.8} = \frac{1}{1} = 1$

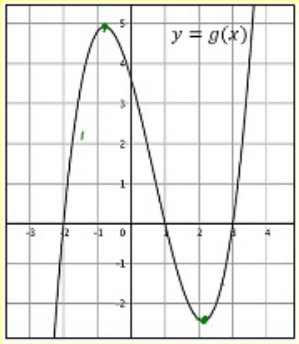
c. $g'(1) \approx \frac{1+1}{0-1.2} = -1.66$

d. $g'(0) \approx \frac{3-2}{0.2-0.6} = \frac{1}{-.4} = -2.5$

Example 4: For which value(s) of x is $g'(x) = 0$?

$x \approx -0.8$

$x \approx 2.1$



Example 5: Put the following values in order from least to greatest.

a. $g'(0), g'(-2), g'(2)$

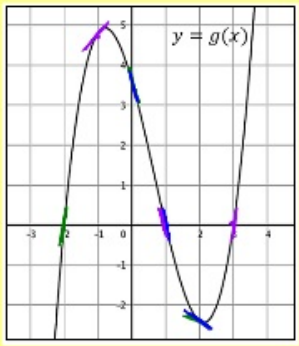
$g'(0) < g'(2) < g'(-2)$

b. $g'(0), g'(1), g'(2)$

$g'(1) < g'(0) < g'(2)$

c. $g'(-1), g'(3), g'(1)$

$g'(1) < g'(-1) < g'(3)$



Approximating a Derivative with a Calculator

Example 6: Find the following values given that $h(x) = \ln(x^3 + 11)$.

a. $h'(-1)$ ≈ 0.300

b. $h'(0)$ ≈ 0

c. $h(4)$ ≈ 4.317

d. $h'(5.1)$ ≈ 0.543

Example 7: Find the following values given that $k(x) = \frac{\sqrt{3x+8}}{1.2x+e^x}$

a. $k'(-2)$ ≈ -0.837

b. $k'(\pi)$ ≈ -0.127

c. $k'(e^2)$ ≈ -0.003

d. $k'(-0.4281)$

≈ -247.735

x	-2	0	1	3
$h(x)$	3	4	-5	0
$h'(x)$	0	2	-4	2

The function $h(x)$ is differentiable for all values of x with selected values of h and h' in the table above.

Example 8: Write the equation of the line tangent to $h(x)$ at $x = 1$.

$h(1) = -5$
$h'(1) = -4$

$y + 5 = -4(x - 1)$

x	-2	0	1	3
$h(x)$	3	4	-5	0
$h'(x)$	0	2	-4	2

The function $h(x)$ is differentiable for all values of x with selected values of h and h' in the table above.

Example 9: Write the equation of the line tangent to $h(x)$ at $x = -2$.

$h(-2) = 3$
$h'(-2) = 0$

$y - 3 = 0(x+2)$

$y - 3 = 0$

$y = 3$

Example 10: Let $k(x) = x^2 + 5x - 3$. Write the equation of the line tangent to $k(x)$ when $x = 3$.

$$k'(x) = \lim_{h \to 0} \frac{k(x+h) - k(x)}{h}$$

$$= \lim_{h \to 0} \frac{(x+h)^2 + 5(x+h) - 3 - (x^2 + 5x - 3)}{h}$$

$$= \lim_{h \to 0} \frac{\cancel{x^2} + 2xh + h^2 + \cancel{5x} + 5h \cancel{-3} \cancel{-x^2} \cancel{-5x} \cancel{+3}}{h}$$

$$= \lim_{h \to 0} \frac{2xh + h^2 + 5h}{h}$$

$$= \lim_{h \to 0} (2x + h + 5)$$

$$= 2x + 5$$

$k(3) = 9 + 15 - 3 = 21$ $\quad EQ_{TAN}: y - 21 = 11(x - 3)$

$k'(3) = 6 + 5 = 11$

Example 11 (Calculator Active): The function p is defined as $p(x) = \ln(x^2 + 7)$. Write the equation of the line tangent to $p(x)$ at $x = e^2$.

$P(e^2) = 4.121$

$P'(e^2) = 0.240$

$y - 4.121 = 0.240(x - e^2)$